

Near Field Communication (NFC)

What is NFC?

Near Field Communication (NFC) is a short-range wireless technology that makes life easier and more convenient for consumers around the world.

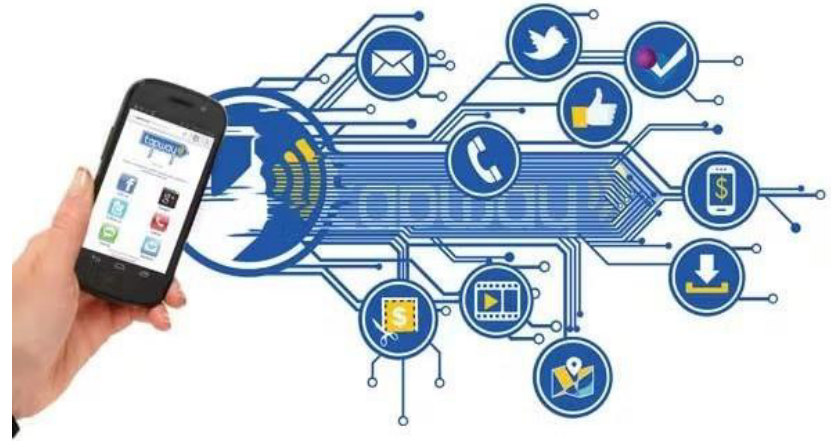
It enables communication between two electronic devices over a distance of 4 cm (about 1.5 inches) or less.

NFC is the technology behind contactless cards, mobile payments, and simple data sharing with just a tap.

Key takeaway: It's all about simple, secure, and intuitive interaction.

Near Field Communication (NFC)

- A short-range wireless connectivity technology mainly aimed for mobile or handheld devices.
- A subset of RFID technology with the exception that NFC is designed for use by devices within close proximity to each other (less than 10cm) thus providing more privacy and security.
- Allows two-way communication between endpoints, whereas RFID system was one-way only
- Operates at 13.56 MHz
- Low Data transfer rate: up to 424 kbits/second thus reducing power consumption.



The Origins of NFC

Roots in RFID: 1983

NFC is built upon Radio-Frequency Identification (RFID) technology. The first patent related to RFID was filed by Charles Walton.

Standardization: 2004

The NFC Forum was established by Nokia, Philips (now NXP), and Sony to advance the technology and ensure interoperability.

The Birth of NFC: 2002

Sony and NXP Semiconductors collaborated to develop the specific technology we now know as NFC.

Comparison with Bluetooth

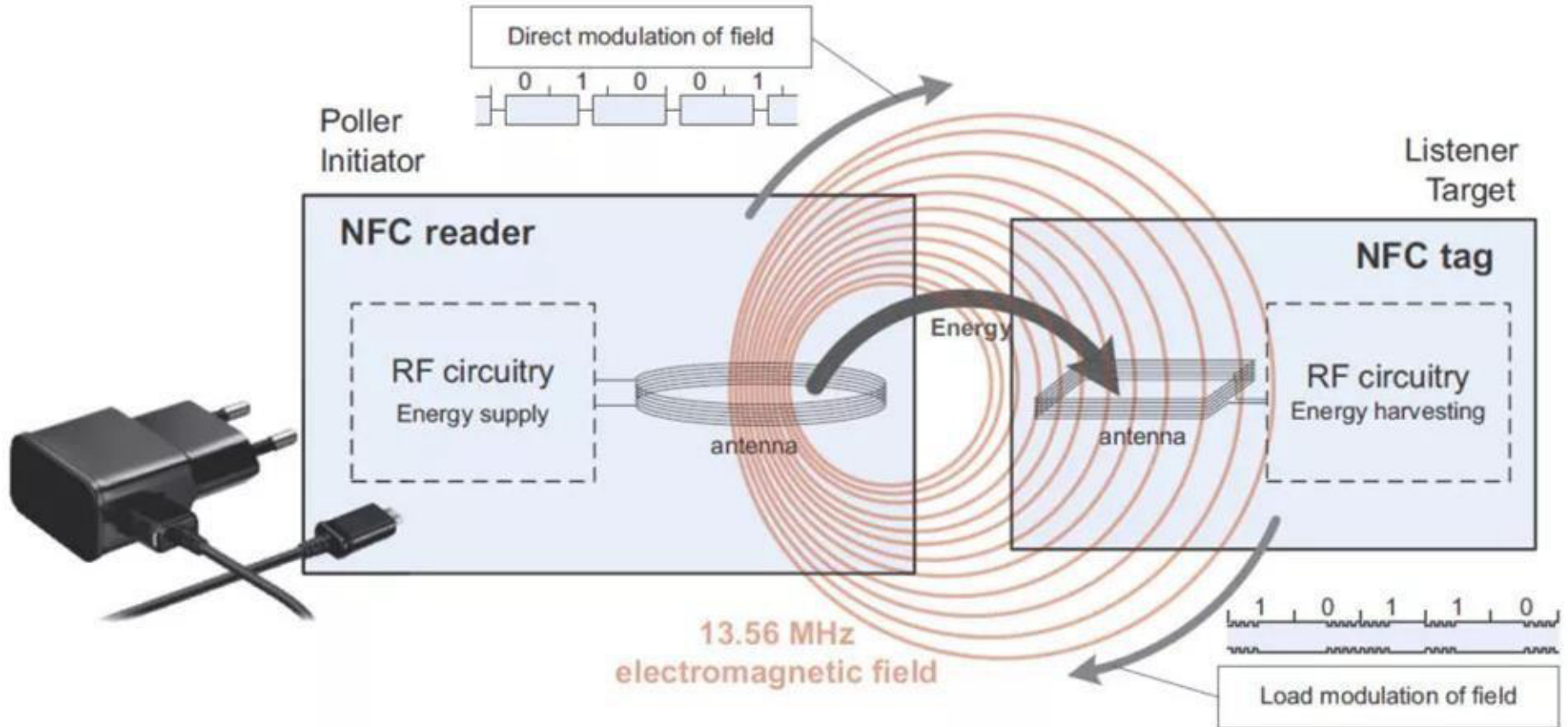
Criteria	NFC	Bluetooth
Power consumption	Low	High
Range	Short (up to 10 cm)	Long (up to 10 meters)
Data transfer speed	Slower (up to 424 kbit/s)	Faster (up to 2.1 Mbit/s for Bluetooth 2.1 and 1 Mbit/s for Bluetooth Low Energy)
Connectivity	Faster (less than 1s to establish a connection)	Slower (may require manual pairing)
Suitable for	Passive devices like tags and advertising signs	File transfers, sharing connections to speakers, and more
Applications	Mobile payments (Samsung Pay, Android Pay, Apple Pay)	File sharing, music streaming, connecting devices

NFC

Applications



Operation of NFC



Operation of NFC (Ctd...)

- The device that generates the electromagnetic field is called poller (initiator), and the second device is the listener/ tag /target.
- This generated field is inductively coupled with the other device in the near field, much like in a transformer.
- Data transmission from poller to listener involves the poller modulating the amplitude of its field, which is registered by the listener.
- The transmission from listener to poller takes place by means of load modulation: The listener can change the amount of drawn energy over time (by changing the impedance). Feedback changes the amplitude of the alternating electromagnetic field, which the poller can register.

NFC's Modes of Operation



Reader/Writer Mode

An NFC device (like a phone) reads or writes data to an NFC tag. **Example:** Tapping a smart poster to open a website.



Peer-to-Peer (P2P) Mode

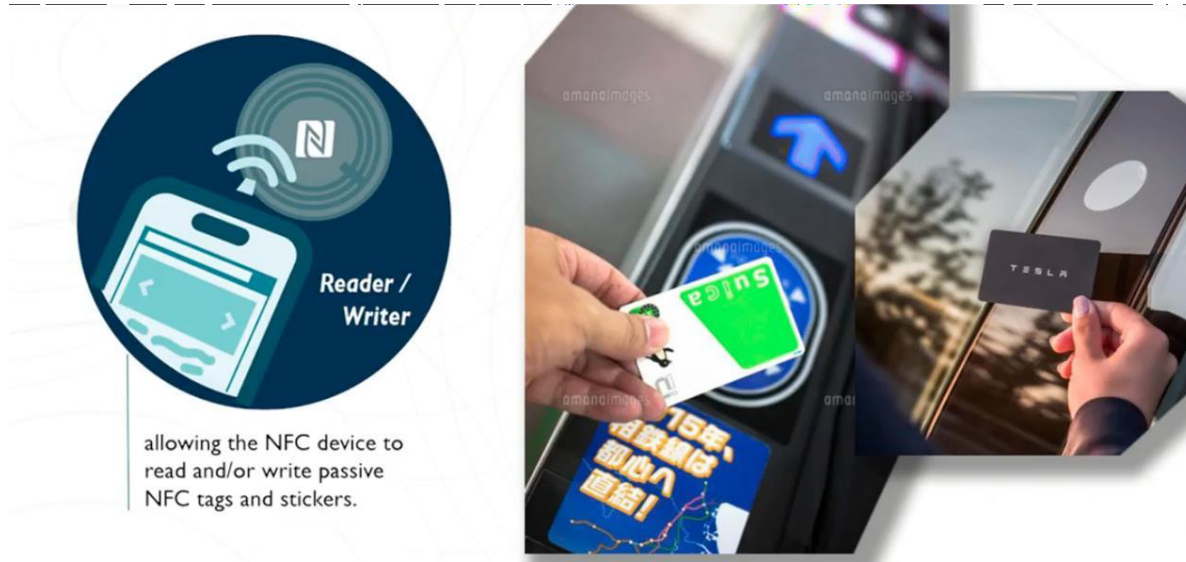
Two NFC-enabled devices exchange data with each other. **Example:** Sharing a photo or contact by tapping two phones together.



Card Emulation Mode

An NFC device acts as a contactless smart card. **Example:** Using your smartphone to pay at a terminal or enter a secured building.

NFC Reader mode



- An NFC device is in "**reader mode**" when it acts as an NFC reader.
- The NFC reader mode allows the NFC device to read or write passive NFC tags and stickers. NFC applications such as transportation or ticketing systems use this mode. Even the key card to open a Tesla's car door operates in this mode.

NFC Card Emulation mode



- The NFC "**card emulation mode**" allows the NFC device itself to act as an NFC card.
- The emulated NFC card can then be accessed by another NFC reader, such as a point-of-sale machine. The best example of this mode is Apple Pay. In this case, the iPhone becomes a contactless credit card, and we can then use it to buy goods in stores

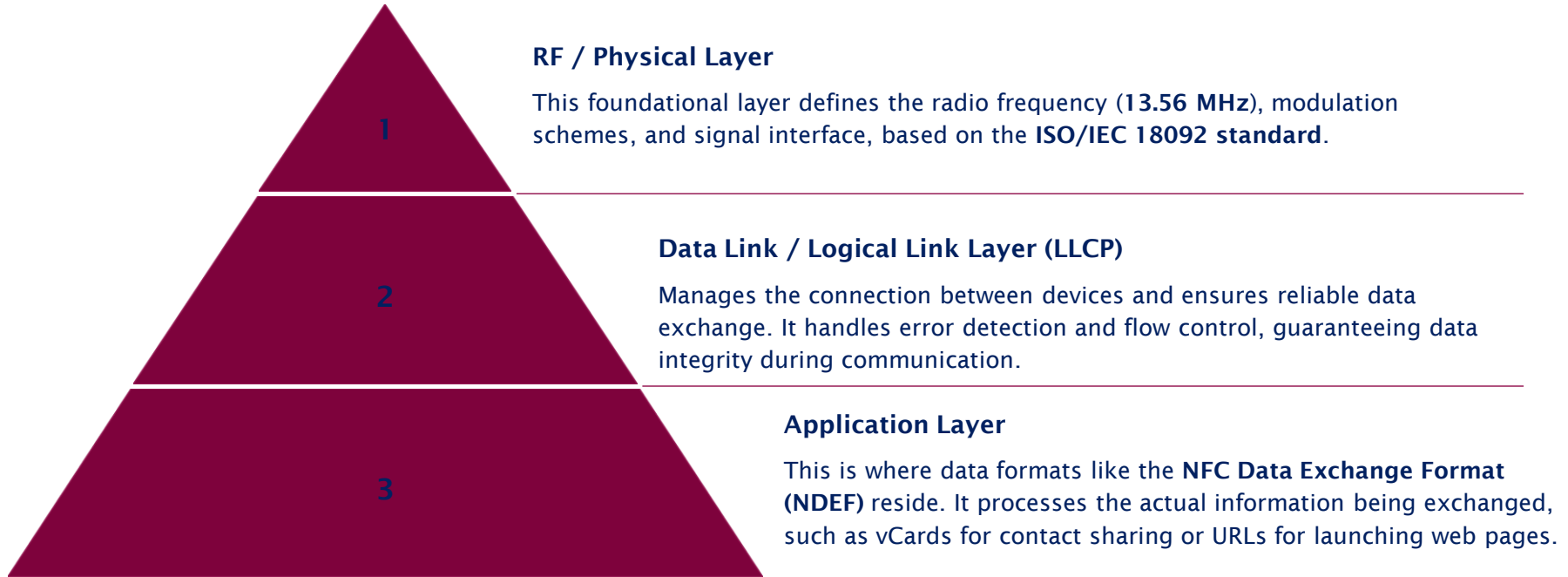
NFC Peer to Peer Communication mode



- Finally is the NFC "**P2P mode**."
- This "peer to peer" mode allows the NFC device to exchange data with another NFC peer. One example of this mode is the Android Beam feature introduced by Google back in 2011. Besides that, this mode is not widely used in the real world.

Layers of Communication

The NFC protocol stack operates in distinct layers, each handling specific aspects of communication. This layered approach ensures interoperability and modularity, similar to the OSI model but simplified for NFC's unique requirements.



Each layer builds upon the one below it, allowing NFC to provide seamless and secure short-range communication for a wide array of applications.

Incorporation of NFC in IoT

- With a straight forward tap-and-go mechanism, NFC makes it simple and intuitive to connect two different IoT devices.
- Because NFC chips must be in close proximity of each other to initiate a transaction, NFC is a clear
 - sign that the user intends to take a certain action. The short range of NFC also protects against
 - unauthorized access by hackers.
- NFC includes built-in features such as encryption that cut down on the potential for eavesdropping and other malicious activities.
- Even objects without power or an IoT connection can passively exchange data via NFC tags. Users
 - with an NFC-enabled device can tap the gadget to get information such as URLs.
- An NFC tag can harvest energy to power other circuits or devices.